Electronic commerce back-office integration

K Lamond and J A Edelheit

This paper discusses some of the lessons learned from integrating two customers' back-office systems with a pilot BT-managed business-to-business electronic catalogue service. An overview of the two pilot implementations describes the experience and offers some important lessons learned.

Three types of lesson learned are presented in this paper — firstly those based on the design and implementation of both pilot catalogue systems, secondly about the importance of proper site security for Internet applications, and lastly about relationship building and information gathering — in the hope that others will be able to use this information in helping to avoid some of the mistakes and pitfalls encountered in this work. As George Santayana once said: 'Those who cannot remember the past are condemned to repeat it.'

1. Introduction

Electronic commerce systems (eCommerce), like any other complex system, are composed of many parts. These parts are the components which handle different processing functions that, when assembled together, provide a fully operational system. Some of these parts are more visible than others, and receive more of our attention when performing research in eCommerce or designing an eCommerce system. These parts have high visibility because they either involve new areas of technology or the users have direct interaction with the part. Some examples of the highly visible parts include components involved in information display, information discovery, and payment handling.

Back office integration, the actual subject of this paper, is not one of the high visibility parts of eCommerce. The need to have two disparate systems communicate with each other has existed for a while. However, just because back-office integration is neither new nor visible to the normal user, does not mean that it is unimportant. Back-office integration plays a vital role in eCommerce, and makes many of the highly visible parts of eCommerce possible.

This paper is not academic. It is based on practical experience with eCommerce and back-office integration. While it was never assumed integrating a company's back office to BT's eCommerce system would be easy, it was never expected to be as challenging as it was. It is intended that this paper communicates the lessons learned from this experience, so others may avoid some of the pitfalls encountered.

A bit of context is necessary before proceeding further. This paper is based on the back-office integration experience from delivering two eCommerce systems to end customers. The type of systems being developed and their operational environments influenced these experiences. Thus a quick background about the environment of these two systems is necessary to provide context for the lessons learned.

Both systems are Web-based catalogue systems for business-to-business trading, and were designed to run as BT-managed services. This means that the eCommerce system is hosted in a BT data centre and must communicate with back-office systems owned, operated, and located on customer premises. Communication between the eCommerce servers and back-office systems took place through the Internet. Furthermore, while BT controlled the managed service environment, it had no control over the back-office environment of the companies with which it was working. Several of the lessons learned focus on how to deal with the IT professionals that maintain these systems.

2. Definitions

Before discussing the detail of the back-office integration, it is necessary to define what back-office integration means in the context of this paper. Simply put, back-office integration enables two or more systems to communicate with each other, and one of these systems is a company's back-office. In terms of BT's eCommerce work, this meant establishing a communications link between BT's eCommerce servers and the pilot participant's back-office machines.

For the purposes of this paper, back-office solutions have been grouped into two types — batch-oriented and real-time. Both methods pass data between an eCommerce and a back-office system. However, the two methods vary greatly in the functionality they provide, and in their complexity to implement.

2.1 Real-time

Real time can be a misleading adjective for any system, because the definition of real-time changes depending on the environment and type of system being discussed.

The term real time evolved from mission-critical systems that required certain actions to take place at specific intervals and in a specific sequence. A system operated in real time if it was fast enough and robust enough to ensure that all actions took place in a specific time interval — usually this time would be measured in fractions of a second after a trigger event occurred.

In the eCommerce world, real time has a slightly different meaning. It describes the ability of an eCommerce system to interact with another system during the order process, either to obtain data that is too volatile to store in an on-line system, retrieve data that is too sensitive to store in the on-line system, or update data in one of the systems based on an event. While this interaction must be fast, multi-second delays are acceptable in an eCommerce environment instead of the fractions of a second required by critical systems.

Real-time integration is difficult to implement and should only be used when customers have requirements that can only be met by using real-time integration. These requirements include:

- dealing with the volatile stock levels of their inventory — quotations even a couple minutes old may be too stale to be useful,
- performing real-time credit checks on corporate accounts — a company may not want to allow someone to make a purchase on faith that they have the credit to cover it, and credit information on customer accounts is considered too sensitive by many companies to allow it to reside on an outside server,
- providing current account and order status — companies can save money if their customers use the catalogue for information for which they would normally call a customer representative,
- running credit authorisations through their back office — some companies may prefer that credit cards be authorised through their own acquirer via their back-office accounting system instead of using an outside system,
- displaying prices that are based on volatile market rates — companies that deal with commodity items or extend loans may tie their rates to market prices at the moment of sale,
- matching services provided by a competitor — a company's customers may start using a competitor's catalogue if it has more functionality.

2.2 Batch-oriented

Batch-oriented denotes the common practice of sending data between two or more machines at set intervals. Data that needs to be sent to update information on another machine is gathered together and held locally until the next scheduled transfer time. Thus, an eCommerce server using a batch-oriented interface stores purchase orders locally, and, at set intervals, forwards the orders to the back-office system.

A similar approach is used to transfer price, product, and stock inventory updates from the back office to the eCommerce server.

Batch-oriented back-office integration normally uses structured text files as the medium for passing data between an eCommerce server and a back office. Structured text files are relatively easy to work with.

Batch-oriented integration should be used whenever a customer does not have requirements that can only be met using a real-time interface. A batch-oriented interface is quicker, easier and less costly to implement.

3. Batch-oriented back-office integration pilot scheme

The first pilot participant was a medium-sized company that manufactured decorating supplies. Both BT and the pilot participant had their own goals for the pilot. The customer wanted an Internet catalogue they could use to provide product and promotional information to their re-sellers, and accept product orders on-line placed through the catalogue. BT wanted to gather requirements for a proposed business-to-business Internet catalogue managed service.

The first eCommerce pilot catalogue was implemented using batch-oriented integration. This was done for two reasons. Firstly, BT's inexperience in catalogue development argued for avoiding the complications of real-time integration. Secondly, the pilot participant had no requirements that made real-time integration necessary. In fact, their IT infrastructure and inventory-tracking system were not sophisticated enough to provide accurate, real-time data about inventory levels and customer orders.

Back-office integration for the first pilot scheme consisted of a system to send files between the eCommerce catalogue and the pilot participant's back office (see Fig 1). The files from the catalogue consisted of purchase order requests from the participant's customers, while the participant's back office sent files containing batch updates of product pricing and order status notifications.

3.1 Issues and lessons learned

While the first pilot never went live, it did progress far enough to provide insight into some back-office integration issues. The back-office integration portion of the pilot was tested successfully before the pilot was ended. Also, most of the issues encountered occurred during the analysis, design and implementation phases of the pilot.

Some of the lessons learned during the first pilot scheme are more applicable to general information gathering, rather than specific batch-oriented back-office solution issues. Information-gathering lessons are documented in section 6.

Standards are not always agreed formats

In eCommerce system development, solutions are always sought that could be reused in later pilots and production systems. Thus, the EDIFact EDI standard was chosen for purchase order messages sent from the catalogue system to the pilot participant's back office.

Naivety led to the belief that EDI messages had strict definitions, and a basic EDI purchase order from one company would be easily read and understood by any EDI-capable supplier. While EDIFact EDI messages contain a structured format, many EDI message definition areas are vague, and these areas must be negotiated between the two parties exchanging the messages. The EDIFact standard also provides many optional fields that may be used to express information about an order; some of these optional fields have almost the exact same meaning. Thus, trading partners must negotiate which fields within a specific EDI message type will be used.

The EDIFact EDI standard describes what message formats and fields an EDI parser needs to handle in order to be EDIFact compliant. While the fields are strictly defined by the standard in the length, type of characters they can contain, and the order in which they appear, only vague guidelines are given to what the actual content of a field should be. Even when the field content is well defined, the meaning of that content is not.

For example, one field that has different meanings is the quantity field. Say a supplier sells item X in packs of six. The supplier may get EDI purchase orders from two buyers, one requesting quantity of 1 of item X and the other requesting quantity of 6 of item X. While the number in the quantity field is different, both buyers are requesting one pack of item X. Because some buyers use the quantity field to state the number of packs of an item they wish to purchase, while others use the quantity field to designate the number of individual units of an item they wish to buy. Thus the supplier has to use prior knowledge about the buyer's EDI message format to determine if quantity 6 means 6 packs of item X or 6 individual units, i.e. one pack of item X.

E-mail is not the simplest transport system

Once it was decided to implement a batch-oriented data transfer approach, it was necessary to choose a method for transporting data between the two machines. The first choice was SMTP (Internet) e-mail, a standard protocol. Since the first pilot participant had a connection to the Internet, it was assumed that they could receive SMTP e-mail. In addition, it was concluded that it would be an easy, reusable solution.

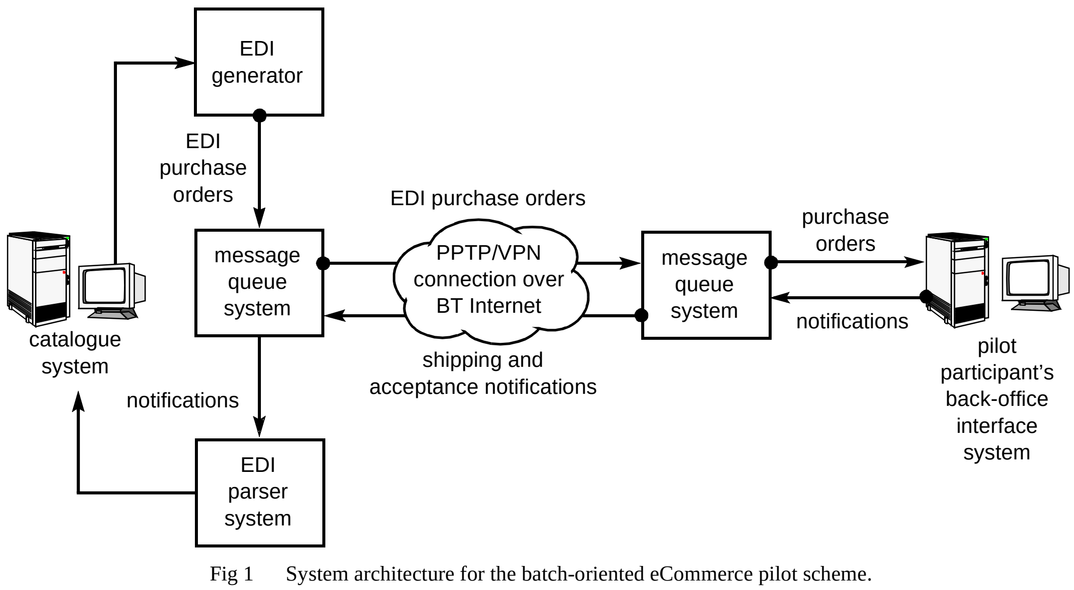

Fig 1 System architecture for the batch-oriented eCommerce pilot scheme.

The more e-mail was considered as a transport mechanism, the less satisfactory it appeared. Several factors about SMTP and the participant's e-mail system caused the abandonment of the use of SMTP e-mail as a transport mechanism. These factors included:

- SMTP-compliant e-mail systems are not as ubiquitous as they seem — most corporations use proprietary LAN-based e-mail systems for sending internal e-mail and use a gateway for sending and receiving SMTP mail,
- since LAN-based e-mail systems and their gateways are proprietary, it would not be possible to reuse a solution tailored around the pilot participant's mail system with companies using a different e-mail system,
- since the pilot participant already had their SMTP gateway listening to the normal TCP/IP port for Internet mail, a specialised SMTP mailer could not be installed on that TCP/IP port,
- SMTP mail does not have a reliable delivery notification mechanism and a 100% fail-safe approach was required.

Abandoning e-mail based transport meant finding a new solution, which leads on to the next lesson learned.

Proprietary can be good, in a closed environment

While SMTP mail was first on the initial list of data transport methods, Microsoft's PPTP protocol was probably last. As PPTP is a proprietary protocol from Microsoft[1], it only runs on Microsoft platforms. Being good system engineers, which means being prejudiced against proprietary protocols, it was initially crossed off the solution list.

At the start of the research to identify available open standards, it was thought that either SMTP or FTP would meet the requirements. After further research, this approach had to be abandoned for two reasons — SMTP lacks reliable delivery notifications and there were security concerns about using FTP. Being unable to find an open protocol that met the requirements, PPTP was revisited. It was quickly concluded that, while PPTP is a proprietary protocol, it was acceptable because the back-office integration part of this pilot was a closed system and did not have to talk to a variety of machines from the outside world. PPTP would only be used to communicate between two NT systems — BT's electronic catalogue and the pilot participant's electronic catalogue-related messaging workstation.

While open protocols are usually preferable, there are situations when a proprietary solution is more appropriate. Closed systems are more suited for proprietary solutions. It is important not to let developer prejudice stop people from looking at what may be the best solution.

Control is good

One fundamental assumption that proved to be true is that every company has a different back-office configuration. This fact presented us with the challenge of how to create a reusable, batch-oriented interface, when each company will have a different back-office environment.

After much contemplation, the most workable solution was for BT to have control of both the hardware and software at the catalogue server and back-office interface. This ensured that there would be at least one machine on the pilot participant's network with which BT could communicate.

The solution was to require the pilot participant to install BT's communications software on a dedicated workstation that they would provide.

4. Real-time back-office integration pilot scheme

The second pilot provided a chance to expand on the lessons learned from the first catalogue system, and to gather requirements for a more complex business-to-business Internet catalogue. The participant in the second pilot was an electronic parts distributor. They saw their competition moving from CD-ROM based catalogues to Internet catalogues for providing product information and allowing customers to place orders. They needed to keep up with their competitors, and also saw the Internet as providing a new channel to market.

The second pilot system was required to be able to support a real-time interface to the participant's back office (see Fig 2). The real-time interface was required for the following reasons:

- the dynamic nature of their inventory stock levels required the electronic catalogue to be able to allocate and deallocate inventory from the back-office system during the order process,
- the pilot participant had a trading model that required that a customer be told if an item was out of stock prior to the order being finalised,

[1] Microsoft has presented PPTP to the IETF and plans to make it into an open protocol.

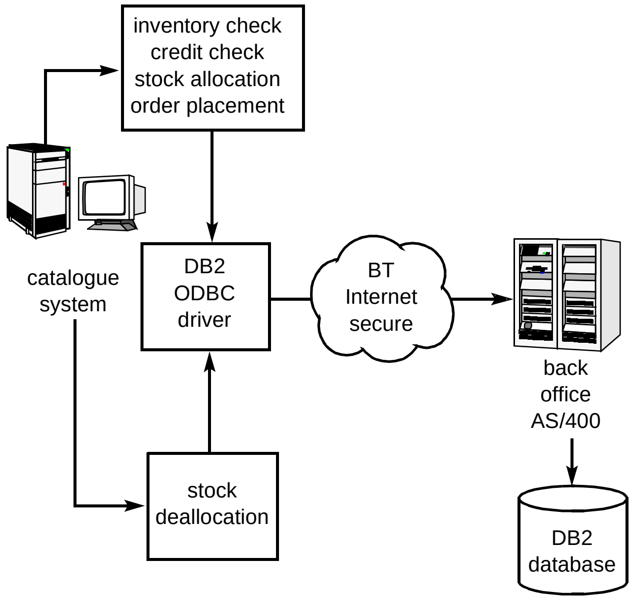

Fig 2 System architecture for real-time eCommerce pilot system.

- the pilot participant's trading model required that the order-entry system confirm that the customer has available credit limit to draw on — the credit checks review both the customer's outstanding balance and any unbilled back orders.

4.1 Issues and lessons learned

Some of the issues encountered during the second pilot are specific to creating a real-time back-office solution for a customer. Other issues relate to information gathering and creating working relationships with the customer and their IT team. Many of the back-office integration issues arose from our working relationship with the pilot participant's contract software developers. While the pilot participant was highly motivated to ensure the success of the pilot, the contractors did not always share this motivation.

This section deals with issues specific to delivering a real-time back-office integration solution. The issues and lessons learned from this second pilot scheme on information gathering and team relationships with the customer will be discussed in section 6.

Clean that data before you bring it in here

At the start of the back-office integration for the second pilot system, problems with dirty data were not anticipated. Since the pilot participant's production system would provide product pricing and other data to the electronic catalogue, it was expected that the data received from the back-office system would be clean and directly usable.

In an AS/400 DB2 environment, the product maintenance data received from the pilot participant may very well be clean. However, BT's Oracle database could not correctly handle the data from the AS/400. It was found that many entries in their DB2 database had leading or trailing white space, and it was impossible for the Oracle database to match it with already existing Oracle data.

The solution to the white space problem with the data was to filter out the extra white space from all data received from their back office. While filtering white space may not be an issue for all implementations, experience suggests that it is prudent for developers to implement filtering code as a standard part of any back-office integration effort.

Test machine? This isn't a military operation!

The heading above is an actual response from the pilot participant when asked about a test environment that could be used for the development and testing of the back-office integration portion of the pilot catalogue system.

It was surprising to find that the pilot participant operated without a back-up system, and that they also carried out development on their operational system. If a newly developed piece of code does not work properly, any errors it creates will show up in their production data. If a developer accidentally deletes necessary data or an important routine, they would be out of business until the problem is fixed. The pilot participant relied on their back-office system for all parts of their operation, from taking orders and accounting, to tracking inventory in their warehouse. They could not afford for something to go wrong with their back-office system.

Fortunately, BT was able to convince them of the need for a test machine for the development and testing of the back-office integration portion of the pilot catalogue. The team felt that the liability and risks that would be presented by interfacing directly to their production machine during development and testing were unacceptable.

While it was surprising to find that they did not have a test system, it seems that only having a production machine is relatively common. Back-office hardware can be expensive and many companies, especially SMEs, see the investment in a development and test environment as an unaffordable luxury. It is important to convince eCommerce customers requiring a real-time solution to invest in a test system for their office. Real-time solutions that involve integration to a customer's back office should not be undertaken without access to a test machine on the customer's site.

Speak AS/400

The IBM AS/400 system seems to be the back-office system of choice for many companies. Having some expertise with AS/400 systems will help in any back-office integration tasks that arise.

Did we say 24 × 7?

A big selling point to on-line catalogues is that they effectively extend the hours a company is open for business. The Web is open 24 hours a day, seven days a week, so an on-line catalogue is available all the time.

Initially, our second pilot participant was adamant that the catalogue system be available 24 hours a day, seven days a week, 365 days a year. They wanted their customers to be able to log on and order at any time. This desire, however, conflicted with their back-office system's operational need to go 'off-line' and into a batch-processing mode each evening. It therefore became necessary to allow users to select but not finalise their orders while the back-office system was in batch-processing mode.

Acceptance testing

A controlled development environment is a wonderful place to acquire a false sense of security about the stability of your system, and find out how smoothly it will interact with the customer's back office once it goes live. Unfortunately, a false sense of security must eventually be dispelled, preferably before the system goes live. This is where acceptance testing plays a role.

Good software development practices dictate several levels of testing before any system is released. Unit testing is done first to find problems within individual components of the system. System integration testing is designed to pinpoint problems with the system as a whole. The tests do fulfil their purpose of catching most problems before the system goes operational. However, since the tests are executed in a controlled environment and are not subject to the unexpected events that occur in a production or live environment, not all problems are found. While developers have a better understanding of a system than the user, users have an uncanny ability of uncovering problems with a system by trying to use the system in ways the developer would never have thought of.

Acceptance testing permits the customer to run the software through its paces and make sure both that it meets their needs, and that it integrates successfully with their environment. For the second pilot system, a three-phase acceptance test was established with the customer. This three-phased approach was adopted for the following reasons:

- it allowed a different aspect of the system to be tested in each phase, thus making it easier to find and correct problems that arose,
- the system could gradually be moved from a highly controlled environment to the actual production environment — variables introduced by new network configurations and running over the Internet were slowly introduced during testing, again making it easier to identify and fix problems as they arose,
- the first two testing phases established the pilot participant's confidence in the system before it was connected to their production system in the last phase.

The three phases were structured as follows:

- the first phase tested functionality of the system that had been developed — this phase did not involve connecting to their back office, but made use of the development AS/400 to simulate their back office,
- the second phase tested the back-office integration logic by connecting to their test AS/400, thus providing a controlled environment on the customer's site, where they had visibility of all the back-office database transactions — this allowed any problems with the back-office integration routines to be captured without endangering their live AS/400,
- the final phase was to test the system against their live AS/400 before the site was announced to the public.

Mining for statistics

One benefit that on-line catalogues provide over paper or CD-ROM catalogues is that on-line catalogues can capture significant amounts of information regarding a user's interaction with the catalogue. A properly designed on-line catalogue will allow data mining for statistics beyond the normal user hits and products purchased. It allows a company to collect information such as which products users often look at but do not purchase, which categories of products are the most and least viewed in the catalogue, and what items users are searching for that may not appear in the catalogue.

The first step in designing a catalogue that maximises its data mining potential is to establish what data about customer catalogue viewing and ordering behaviour will aid the seller's marketing efforts. What information does the seller want to learn about the users of its on-line catalogue? Is the company interested in profiling all the products looked at by a customer before they decide on what to purchase? The seller may want to see what products are being viewed by a large number of users but not being purchased, and then create a promotional scheme around that product to see if it stimulates sales. Alternatively, they may want to create personalised mailings to users based on their buying and browsing habits. Once the data mining requirements are known, they can drive the design of catalogue data mining enhancements.

With the second pilot catalogue, the mistake was made of not gathering specific data mining requirements during discussions with the participant and thus customised data mining was not taken into account when the system was designed. Now that the system is live, there is information that it would be useful to retrieve about catalogue usage but is not accessible, because the system was not designed to readily capture this data.

One important note on capturing data mining requirements and on delivering data mining reports to the seller is to educate them on what information can be mined and how to read the related reports. The main reason detailed data mining requirements were not captured from the pilot participant is that they did not know what information they wanted. Users must also be educated about how to read site usage reports because they contain many statistics including bandwidth usage, site hits, and user sessions. The customer needs to be taught what each of these items means, and which items provide the most useful information. For example, customers may not understand that the number of user sessions is a more important statistic than site hits, as it gives a better picture of how many people are accessing the system.

5. Site security

Security is an important part of projects involving the integration of two machines that reside on the Internet. Any time you connect to a customer's back office over the Internet, you are putting their back office at risk from outside intrusion. For example, if the catalogue server is compromised, the back office system is threatened. While the greatest risks are posed by systems supporting real-time back-office integration, systems using a batch-oriented approach to integration are not risk free. For example, the flat files that are used to pass data between machines in a batch-oriented system could become compromised, and used to pass false orders or pass incorrect data to the customer's back office.

It is easy to dismiss the calls for proper security. It is easy to believe that a project is too small or not visible enough to draw the interest of hackers. However, experience has shown that hackers **will** try to break into your system on the Internet! Firstly, within one hour of the pilot site becoming operational, the site was probed from the outside for security weakness. This means that hackers knew about the site almost the instant it became live. Secondly, the Web logs for the pilot site usually contain 10 to 20 unusual user fingerprints, fingerprints that come from untraceable accounts, accounts that are often used to hide a hacker's identity. Lastly, there is proof that a real hacker tried to break into the on-line catalogue.

The failed break-in took place at 4:00 am GMT, 7 September 1998. In reviewing system logs, six specific attempts to access the catalogue using commonly known security holes were identified. Hacking is a real threat — the pilot site has low visibility, yet it was targeted as part of an organised hacking campaign.

Outlined below are important security measures that have been taken to secure the pilot catalogue system, and should be part of the operation of any live eCommerce system:

- all software on the machine, including the operating system and communications, must be kept up to date with the latest vendor patches — hackers know the security holes that exist in unpatched software and will exploit them,
- packet filtering and killing malformed TCP/IP packets at either a router or firewall should protect the back office — sending a SYN/Flood attack to an AS/400 would cause a complete shutdown of the back-office system,
- software should be installed that keeps track of the version number, file size, and date-stamp of all files on the system, because when hackers break into a site, they will create backdoors to make future access easier by substituting one of their files for a commonly used file on the system — there are a number of commercial products that perform this function,
- audit logs of all activity on the site should be collected and analysed, and then archived off-line — hackers will try to delete any evidence left of their visit on a site,
- all components of the system should be tested, including firewalls and routers, for resilience to outside attacks,
- as overloading services is a classic attack used by hackers, all components should be tested by pushing them to their limits (components should fail gracefully when overloaded) — hackers exploit common failings of overloaded components such as buffer overflows.

As this experience has shown, hacking is a real threat to any system put on the Internet. It is critical to take steps to protect customers from outside intrusion. The steps above provide a good starting point for securing a site that is on the Internet.

6. Information gathering and relationship building

It was debated as to whether this section belonged in a paper about back-office integration. However, discussions about the lessons learned from the two pilot sites, with regard to back-office integration, continually led into issues that were focused on information gathering and relationship building. The information gathered and relationships built with customers, and their representatives, were essential to the success of the back-office integration efforts.

6.1 I think it works like this

One of the hardest tasks involved in integrating a customer's back-office system with a managed eCommerce system is obtaining accurate information on the capabilities of and interfaces into the customer's back office. Fully defined, detailed requirements are difficult to find.

Obtaining extensive information on the customer's back-office system is difficult for several reasons.

- Information

  The designated main contact at the customer's site may not have detailed knowledge of their back-office system. The main contact with the second pilot participant could provide a general overview of the system, but there were many questions he could not answer. There were also several occasions that the information he provided was found to be incorrect.

- Communication

  People who understand the back-office system may not be good at explaining it to others. The job of maintaining a back-office system does not require good communications skills. Information received from the people who maintained the pilot participant's systems was usually terse and incomplete, causing constant referral back to them to elaborate on their descriptions of the system.

- Commitment

  Employees and contractors who manage the back office may be either afraid of the project or just apathetic to it. With the first pilot participant, anything related to back-office integration went on the back burner. There was a sense of apathy surrounding the project. The second pilot participant contracts out the maintenance of their back-office system. At times, it seemed the third-party contractor saw the pilot system and BT as a threat to the work that they were doing.

The only solution to the problems outlined above is to practice patience and persistence. Patience is needed to maintain a cordial relationship with all parties involved, while waiting for more information to arrive or in trying to decipher the information that is sent. Persistence is needed to ensure that the necessary information does arrive. While the customer project leader may be committed to the project, the people who support their back office, the ones from whom you need to get the information, may not be. Persistence is also needed to remind them repeatedly that you are waiting for information, and then making sure that they fully explain what is essential. Patience and persistence are the only things that will get you through a question and answer session about the customer's back office, with someone who does not want to be involved and lacks good communication skills.

6.2 Customer motivation

One of the lessons learned from the first pilot scheme is probably applicable to any pilot effort, no matter what technology is on trial. If the participant has little invested in the pilot, they have little motivation to see the pilot through, and will not provide the resources necessary to make it a success.

The first pilot participant's investment to the on-line catalogue pilot was minimal. They needed to spend some time to cleanse the source data, develop a small amount of software for their back office, and dedicate an NT machine to the project. The data cleansing was something they would need to do eventually, the development was minimal, and the NT machine used for the project was one of their spares. Thus, they had a minimal investment in the pilot and little commitment to see it through.

A junior person in their IT staff was assigned to help, one with inadequate experience for the task. Getting clean data and data updates from them was a long and arduous process. When problems arose, they quickly lost interest.

They probably started out with good intentions. However, since they made no investment in the pilot, it was easy for pilot-related tasks to be put last in the priority list, and thus never get accomplished. In contrast to the first pilot participant, the second participant made a large investment towards their pilot, and was highly motivated. A corollary emerged from our experiences with the two companies. The amount of money invested by a pilot participant dictates the amount of interest and commitment they have in the scheme.

6.3 Relationships, relationships, relationships

The real estate industry has an axiom about the three essential items that make a property sell — location, location, and location. A similar axiom exists for gathering information — relationships, relationships, and relationships.

Obtaining information about a customer's back office is not easy. It usually involves several people and can include third parties the customer hires to maintain their system. It involves working with people who are already stressed by their daily workload and see the project as another item added to an already long list of tasks. It means working with people who are not used to dealing with outsiders about the system on which they work. It involves getting people to help in a task they may not want to do. Building relationships is the key to getting information and help from these people.

Establishing a good working relationship with the people you need to talk to in the information-gathering process makes the process much easier. It was found that after a face-to-face meeting with someone, they were more open and willing to be co-operative. Establishing a personal relationship helps counteract the other influences on a person's time that prevents them from providing the information needed. It is easy to ignore a faceless person; it is not as easy to ignore someone you have met.

6.4 Face time

Face-to-face meetings are critical in establishing a good relationship with the customer and their representatives. It helps build trust between the customer and the development team. Face-to-face meetings have a better flow to them than phone conversations. Another factor is that only 20% of communications are verbal. Humans rely on sight cues like body language to gauge people's responses, and these are unavailable through the telephone. We trust our impressions of a person much more when we can observe the other person's body language. Once face-to-face contact has been made, the image of that person is associated with their voice over the telephone or the e-mail they send. Rapport that is established during face-to-face meetings carries over to communication via other mediums.

The task of gathering information is one of talking to people. It cannot solely be done using a telephone or electronic mail. Trust and rapport is initially built through face-to-face contact, not telephone calls.

6.5 Expect the unexpected

Both pilot schemes were started with preconceived notions about their back-office environments. The first pilot participant was expected to have had a much more sophisticated back-office system than they did, and it was a surprise to find out that they were unable to reliably track their inventory level. The fact that the second participant did not have a test or back-up system to support their operation was also unexpected.

Even where there were no preconceived notions, the details of both back-office systems were disconcerting. The first participant caused surprise with the ever-moving pack unit number in their product data. The table used in their system to hold product data was created without a field to hold the number of units in a pack of a product. To fix the situation, they inserted the number of units in a pack in any of the product description fields that had extra space available. Though the second participant's back office is more sophisticated, it also still managed to spring some surprises. The complexity of the algorithms to perform routine tasks on their back-office system was an example of this, as instanced by the verifying of an account for available credit and the checking of the status of back-ordered items.

One thing that both back offices have in common is that they are 'home-grown' solutions. They were built to meet specific needs the company had at the time of their conception, and then modified as the company's requirements of their back-office system changed. This type of development led to the first pilot participant's wandering pack units, and to the convoluted nature of the algorithms to extract data from the second participant's system.

Beware of preconceived notions — behind every customer's back office will be at least one thing that surprises you and catches you unprepared. Perhaps the only thing that can be expected is the unexpected.

7. Conclusions

It is hoped that the lessons discussed in this paper from the two pilots will aid readers in their back-office integration efforts.

Both pilot schemes provided warning examples about back-office integration. The first pilot application provided lessons about EDI messaging and the creation of batch-oriented solutions. Important lessons about 'dirty data' and customer motivation were also learnt. The second pilot system warned about the trials and tribulations of interfacing with a back-office system in real time.

Some of the most important lessons learned, and hardest to express on paper, were those about gathering information and building good relationships with the customer and their representatives. While the lessons listed in this area can provide guidance to others, they really can only be learned spending time in front of customers. People skills take practice, and, unfortunately, do not always come naturally to developers. However, it is important for developers to talk with the customer. Asking important questions through a third party, who does not understand the questions or issues, is a long and cumbersome process.

The last lesson is that there is much more still to be learned about back-office integration. Each company BT talks to about eCommerce has different requirements, and these new requirements put new demands on any back-office solution. Each company's back-office system uses different hardware, software, or both. Back-office integration is an area full of challenges.

Keith Lamond has worked for Concert Management Services and BT North America Inc since 1994. During that time, he has been involved in both Concert and BT messaging services and represented BT in the applications area at several IETF conferences.

His experience in electronic commerce includes acting as the main system architect and team leader in the design and implementation of two BT eCommerce pilot schemes.

He was awarded his MSIS in 1997 from George Mason University, Fairfax, Virginia, and his BA in 1990 from Mary Washington College, Fredericksburg, Virginia.

Jeffrey Edelheit has worked for Concert Management Services and BT North America Inc since 1995. In his role as electronic commerce development manager for Concert's Network and Systems Engineering, he led a team that focused on electronic commerce applications, research, and standards for both Concert and BT. He was awarded his MSIS in 1990 from George Mason University, Fairfax, Virginia, and his BBA in 1971 from the University of Toledo, Ohio. Prior to joining BTNA, he was a Senior IT Security Engineer for the MITRE Corporation, a US Department of Defense Federally Funded Research and Development Center. While at MITRE, he provided security engineering support for the Joint Chiefs of Staff, the National Security Agency, and the Defense Advanced Research Projects Agency.